

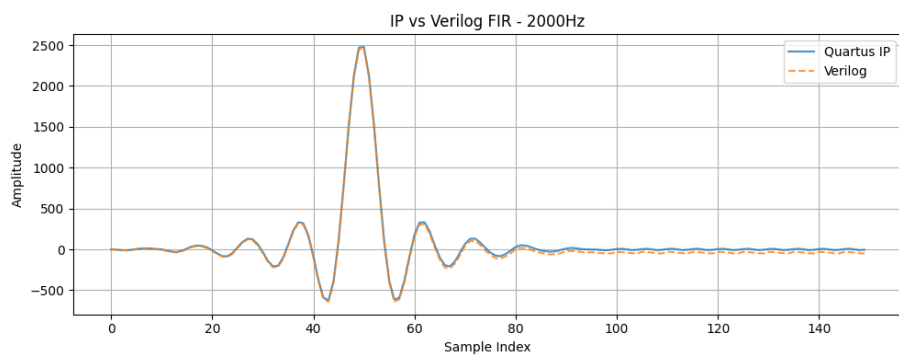
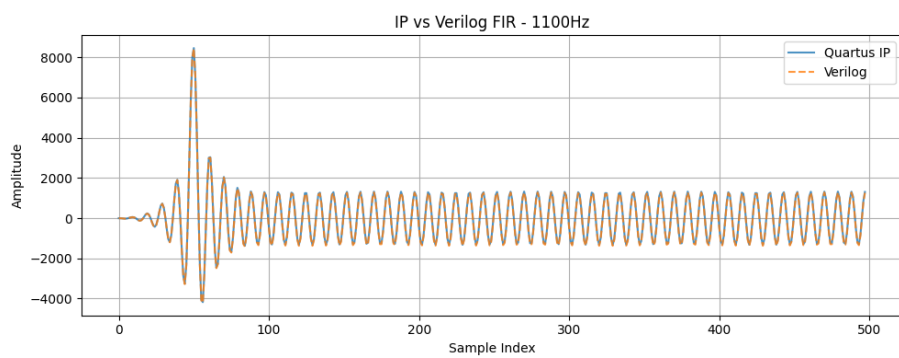
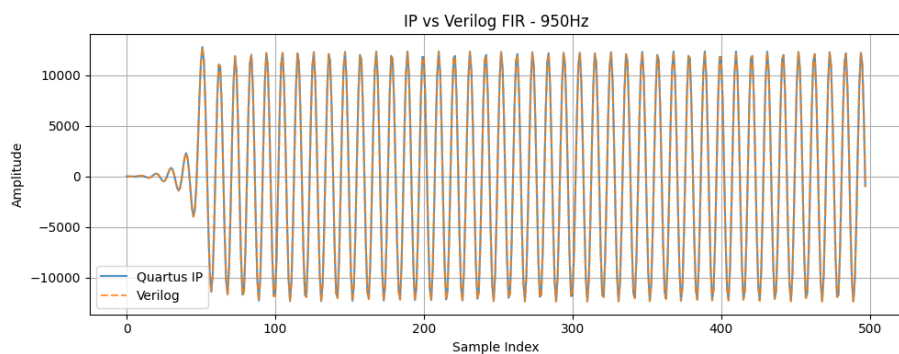


Table 1: Performance and Resource Utilization Comparison

Architecture	F_{max} (MHz)	Slack (ns)	Throughput (MS/s)	Resource Utilization		
				ALMs	Regs	DSPs
Quartus IP	180.18	3.07	180.18	1445	2863	25
<i>Non-Pipelined Implementations</i>						
Direct	44.86	2.707	44.86	1064	1727	25
Optimized	46.82	3.76	46.82	1075	2727	25
Genvar	51.86	5.704	51.86	4099	1872	0 (logic)
<i>Pipelined Implementations</i>						
Direct	191.02	19.198	191.02	1075	2785	25
Optimized	195.82	19.765	195.82	1572	2785	25
Genvar	221.04	20.476	221.04	1572	3173	25